

number of cores available on the machine. That too is done by emulating the extra cores.

- ▶ The hard disk size the virtual machine will have access to: Assuming you allot 20 GB of hard disk space to the virtual machine, you can think of this as “attaching a hard disk whose full size capacity is 20 GB”. Usually the size is allocated on your machine as a file and you can decide the interface. Since everything in there is virtual, you can also create more than one hard disk (unlike partitions on a real hard disk, you would be actually emulating more than one hard disk).
- ▶ The number of network adapters the virtual machine will have and their configuration which determine factors such as:
 - Will the machine be isolated?
 - Will it be using a Network Address Translation (NAT) mechanism to allow the internet access to the virtual machine?
 - Will it be an internal network which can be used only within the various virtual machines running on that virtualization software?
 - Should the Network Adapter appear as if the cable has been disconnected?
 - What should be the MAC address of the network adapter(s)?
 - What type of network adapter it would be?

Depending on the virtualization software, a few more options are provided to help you customise the networking behavior of the operating system running within.

- ▶ The number of removable devices you’ll have in your system.
- ▶ Should it be an ISO file which looks like a DVD drive to the machine or should the machine have access to your real drives?
- ▶ Should the virtual machine be able to access your USB drives?
- ▶ Whether the virtual machine can play audio in the real system.

There are still a few more options which can be configured for every virtual machine you create using the virtualization software. We’ll look into that later as we discuss the various software available for the purpose and their features.

Depending on the options you’ve selected for the virtual machine, the virtualization software will decide what hardware to emulate and how to emulate. A small diagram would help you understand how it all works.

Role of hardware features

Although virtualization is all about fooling an operating system about the

